Volcano – Simple English Wikipedia, the free encyclopedia

Source: https://simple.wikipedia.org/w/index.php?title=Volcano&oldid=10961960, retrieved 2026-10-18.
Licence: CC BY-SA 4.0. Authors: https://simple.wikipedia.org/w/index.php?title=Volcano&action=history. Content unchanged, site layout left out.

A **volcano** is a mountain that has lava (hot, liquid rock) coming out from a magma chamber under the ground, or did have in the past. Volcanoes[1] are formed by the movement of tectonic plates.

Volcanoes can also form where there is stretching and thinning of the crust's plates, e.g., in the East African Rift.[2] Volcanoes are not typically found where two tectonic plates slide past one another.

Volcanism away from plate boundaries is caused by mantle plumes. These so-called "hotspots", for example Hawaii, are thought to arise from upwelling magma from the core–mantle boundary, 3,000 km deep in the Earth.

Most volcanoes have a volcanic crater at the top. When a volcano is active, materials come out of it. The materials include lava, steam, gaseous sulfur compounds, ash and broken rock pieces.

When there is enough pressure, the volcano erupts. Some volcanic eruptions blow off the top of the volcano. Sometimes, the magma comes out quickly or slowly depending on the eruption. Some eruptions come out at a side instead of the top.

Volcanoes are found on planets other than Earth. An example is Olympus Mons on Mars.

Volcanologists are scientists who study volcanoes using methods from geology, chemistry, geography, mineralogy, physics and sociology.

The world's biggest volcano is named Mauna Loa in Hawaii. Mauna Loa is part of the five volcanoes on Hawaii's 'Big Island'. The most recent time this volcano erupted was in 1984. It erupted 33 times in the last 170 years. Like all the other Hawaiian volcanoes, Mauna Loa was created by the movement of the Pacific tectonic plate which moved over the Hawaii hotspot in the Earth's mantle. Mauna Loa is 4,196 meters tall. It is a shield volcano. The largest recent eruption from Mauna Loa left a lava trail 51 kilometres (32 miles) long.

Types of volcanoes

The lava and pyroclastic material (clouds of ash, lava fragments and vapor) that comes out from volcanoes can make many different kinds of land shapes. There are two basic kinds of volcanoes.

Shield volcanoes

These volcanoes are formed by runny, low-silica, mafic lava.

Shield volcanoes are built out of layers of lava from continual eruptions (without explosions). Due to the lava's low viscosity, it spreads out, often over a wide area, forming the shield volcano's characteristic gently sloping sides. The base of the volcano increases over successive eruptions.

Although their sides are typically less steep, shield volcanoes rank among the largest in the world. Mauna Kea, in Hawaii, is the biggest mountain on Earth when measured from its base on the ocean floor.[3]

Stratovolcanoes

A stratovolcano, also known as a **composite volcano**,[4] is a tall, conical volcano. It is built up of many layers of hardened lava, tephra, pumice, and volcanic ash.

Unlike shield volcanoes, stratovolcanoes have a steep profile and periodic eruptions. The lava that flows from stratovolcanoes cools and hardens before spreading far. It is sticky, that is, it has high viscosity. The magma forming this lava is often felsic, with high-to-intermediate levels of silica, and less mafic magma. Big felsic lava flows are uncommon, but have travelled as far as 15 km (9.3 mi).[3] [5]

Two famous stratovolcanoes are Japan's Mount Fuji, and Vesuvius. Both have big bases and steep sides that get steeper and steeper as it goes near the top. Vesuvius is famous for its destruction of the towns Pompeii and Herculaneum in 79 AD, killing thousands.

Caldera

Main article: Caldera

A caldera is a basin-like feature formed by collapse of land after a volcanic eruption. This happens after a huge stratovolcano blows its top off. The base of the crater then sinks, leaving a caldera where the top of the volcano was before. Krakatoa, best known for its catastrophic eruption in 1883, is much smaller now.[3]

How volcanoes are formed

There are two main processes.

Volcanoes are made when two tectonic plates come together. When these two plates meet, one of them (usually the oceanic plate) goes under the continental plate. This is the process of subduction. Afterwards, it melts and makes magma (inside the magma chamber), and the pressure builds up until the magma bursts through the Earth's crust.

The second way is when a tectonic plate moves over a hot spot in the Earth's crust. The hot spot works its way through the crust until it breaks through. The caldera of Yellowstone Park was formed in that way; so were the Hawaiian Islands.

Classification

A traditional way to classify or identify volcanoes is by its pattern of eruptions. Those volcanoes which may erupt again at any time are called active. Those that are now quiet are called dormant (inactive). Those volcanoes which have not erupted in historical times are called extinct.

Active

Main article: List of active volcanoes

An active volcano is currently erupting, or it has erupted in the last 10,000 years. An example of an active volcano is Mount St. Helens in the United States (US).[6]

Dormant (inactive)

A dormant volcano is "sleeping," but it could awaken in the future. Mount Rainier in the United States is considered dormant.[6]

Extinct (dead volcano)

Main article: List of extinct volcanoes

An extinct volcano has not erupted in the past 10,000 years.[6] Edinburgh Castle in Scotland sits on top of an extinct volcano.[7]

Largest volcano on Earth

The Earth's largest volcano has been discovered.[8][9] It is 2 km below the sea on an underwater plateau known as the Shatsky Rise. This is about 1,600 km east of Japan. The previous record-holder, Mauna Loa in Hawaii, is still the largest volcano on land.

The 310,000 km^2 (119,000 sq mi) volcano, Tamu Massif, is comparable in size to Mars' vast Olympus Mons volcano, which is the largest known volcano in the Solar System. It was formed about 145 million years ago when massive lava flows erupted from the centre of the volcano to form a broad, shield-like feature. That suggests the volcano produced a flood basalt eruption.

The Tamu Massif extends some 30 km (18 miles) into the Earth's crust. The researchers doubted the submerged volcano's peak ever rose above sea level during its lifetime and say it is unlikely to erupt again.

> "The bottom line is that we think that Tamu Massif was built in a short (geologically speaking) time of one to several million years and it has been extinct since", co-author William Sager of the University of Houston told the AFP news agency.
> "There were lots of oceanic plateaus (that) erupted during the Cretaceous period (145-65 million years ago) but we don't see them since. Scientists would like to know why... The biggest oceanic plateau is Ontong Java Plateau, near the equator in the Pacific, east of the Solomon Islands. It is much bigger than Tamu – it's the size of France".[8]

Related pages

- List of volcanoes

References

1. ↑ The plural of volcano can be *either* volcanos or volcanoes. Both are correct and it is not a matter of British vs US spelling. *Oxford English Dictionary*.
2. ↑ *Foulger, Gillian R. (2010). Plates vs. Plumes: a geological controversy. Wiley-Blackwell. ISBN 978-1-4051-6148-0.*
3. ↑ 3.0 3.1 3.2 *Earth Science. Austin, Texas: Holt, Rinehart and Winston. 2001. ISBN 0-03-055667-8.*
4. ↑ *"Principal types of volcanoes. USGS". Archived from the original on 2013-05-09. Retrieved 2013-07-02.*
5. ↑ *"Garibaldi volcanic belt: Garibaldi Lake volcanic field". Catalogue of Canadian volcanoes. Geological Survey of Canada. 2009-04-01. Archived from the original on 2009-06-26. Retrieved 2010-06-27.*
6. ↑ 6.0 6.1 6.2 *Ball, Jessica (September 8, 2010). "Voices: Dead or alive ... or neither? Why a dormant volcano is not a dead one". Earth Magazine. Archived from the original on 28 January 2013. Retrieved 14 August 2012.*

7. ↑ University of Edinburgh, "Holyrood Park Geology" Archived 2007-08-30 at the Wayback Machine; retrieved 2012-8-2.
8. ↑ 8.0 8.1 World's largest volcano discovered beneath Pacific. *BBC Science & Environment*. [1] Archived 2019-12-19 at the Wayback Machine
9. ↑ Witze, Alexandra 2013. Underwater volcano is Earth's biggest: Tamu Massif rivals the size of Olympus Mons on Mars. [2] Archived 2014-07-06 at the Wayback Machine

Other websites

Wikimedia Commons has media related to ***Volcanoes***.

- The United States Geological Service (USGS) Volcano page Archived 2004-06-03 at the Wayback Machine
- Glossary of Volcanic Terms from USGS Archived 2007-09-22 at the Wayback Machine
- Volcanic and Geologic Terms Archived 2007-09-14 at the Wayback Machine from Volcano World Archived 2004-08-05 at the Wayback Machine - University of North Dakota (UND)
- Television program (BBC) on the prediction of Popocatepetl's 2000 eruption